

CE 774: Traffic Management and Design

Coursework Project Report

Topic: Sensitivity Analysis at the H-10 Junction, IIT Bombay

submitted by:

Abhijeet Kumar Gaurav, 24M0553
Lekshmi Murlidharan, 24M0544
Deep Medatwal, 210040044
Deven Ratnaparkhi, 210040045
Harendra Pratap Singh Baghel, 210040054
Kartik Khandelwal, 210040072
Mariyam Saqib, 210040090
Mohak Gupta, 210040097

Supervisor:

Professor Archak Mittal

TA Guides:

Aiswaryabsree, Dhaval Barot, Manju Putra Pratyush, Shyam Manchhani, Vaishnavi Shinde



Department of Civil Engineering

Indian Institute of Technology Bombay

Mumbai, 400076, India

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Abstract

Rapid infrastructural development within the IIT Bombay (IITB) campus, including new hostel construction and drainage works, has increased vehicular activity and reduced effective road space near the H10 intersection, causing operational inefficiencies. This study evaluates the current and projected traffic performance of the unsignalised H10 T-junction using microsimulation modelling in PTV VISSIM. A baseline model, calibrated to reflect existing traffic conditions, was used to simulate future scenarios with 2%, 5%, and 7% traffic increments. Sensitivity analysis assessed the intersection's performance in terms of travel time, vehicle delay, queue length, fuel consumption, and Level of Service (LOS). The base scenario indicated severe congestion, with movements from the Energy Building side operating at LOS F and experiencing high delays and queue lengths. Two improvement strategies were evaluated: widening of approach lanes and the introduction of dedicated cycle tracks. Lane widening reduced average travel times by 34 to 50%, delays by up to 75%, and fuel consumption by 70 to 80%. Dedicated cycle lanes dramatically reduced delays and queues for multiple movements, yielding fuel savings of nearly 99% for some approaches. Signalisation was ruled out per IRC:93 guidelines, as traffic volumes and operational characteristics did not warrant a traffic signal. The study concludes that targeted non-signalised improvements, particularly geometric modification and modal segregation, can significantly enhance the operational efficiency and safety of the H10 intersection, offering a practical framework for future campus mobility planning.

Keywords: PTV VISSIM Microsimulation; Unsignalised Intersection; Sensitivity Analysis; Level of Service; Lane Widening; Dedicated Cycle Tracks; Campus Traffic Planning

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1 Introduction

1.1 Background

Urban campuses such as IIT Bombay (IITB) experience evolving and dynamic traffic patterns due to continuous infrastructural expansion and growing student and staff populations. Ongoing construction activities, including new hostel blocks and drainage works near Hostel 10 (H10), are expected to substantially increase vehicular and pedestrian movement in and around the primary corridors of the campus.

Intersections are critical points within any traffic network, where multiple streams of traffic meet and interact. Unsignalised intersections depend heavily on driver behaviour, sight distance, and negotiated priority. Without proper assessment and management, such intersections are vulnerable to congestion, operational delays, and safety risks.

One such important location on the IIT Bombay campus is the H10 T-junction, where routes from Hostel 10, the main academic area, and residential sectors converge. Given the anticipated rise in traffic demand, there is a need to systematically evaluate the operational performance of this intersection.

1.2 Motivation

The H10 intersection is a vital connection point linking hostels, academic buildings, and administrative areas. Several factors currently challenge its operational efficiency:

- **Proximity to key landmarks:** the intersection lies adjacent to KV School and the Infinity Corridor entrance, both contributing high pedestrian and vehicular flows.
- **Rising pedestrian movements:** increased entrances to Hostel 10 have raised footfall, including young students, emphasising the need for pedestrian safety.
- **Traffic and visibility challenges:**
 - Construction activities have resulted in a higher influx of heavy commercial vehicles (HCVs) and auto-rickshaws.
 - Roadway space has been reduced due to ongoing drainage works, affecting the effective carriageway width.
 - The existing Hostel 10 building obstructs sight distances at the junction, complicating manoeuvring for both vehicles and pedestrians.
 - Planned new hostels (H19, H21, etc.) are likely to further escalate traffic volumes in the near future.

Given these challenges, it becomes imperative to evaluate existing traffic conditions at the H10 intersection and explore potential improvement strategies. Using microsimulation tools like

PTV VISSIM, various operational strategies, such as road layout changes, pedestrian crossing improvements, and possible geometric redesigns, can be tested in a virtual environment before implementation. This approach ensures evidence-based planning and enhances both vehicular and pedestrian safety within the campus.

1.3 Problem Statement

The ongoing infrastructural developments within the IIT Bombay campus have significantly altered existing traffic dynamics around the H10 intersection. The main challenges observed are:

- Increased vehicular movement, particularly of autos and heavy commercial vehicles (HCVs), related to construction activities.
- Reduction in effective road width due to drainage construction, leading to restricted vehicular movement space.
- Emergence of localised congestion during peak hours, caused by both the increased volume and reduced carriageway width.
- Limited sight distance near the H10 junction, making vehicle turning manoeuvres riskier and slowing down overall traffic movement.

Combined, these factors contribute to operational inefficiencies at the unsignalised H10 intersection. Thus, a detailed operational analysis is necessary to assess current traffic performance under constrained roadway conditions, simulate future scenarios considering ongoing and planned developments, and identify and recommend feasible improvements to enhance intersection performance and safety.

1.4 Objectives

The primary objectives of this study are:

1. To develop a calibrated microsimulation model of the H10 unsignalised intersection using PTV VISSIM.
2. To evaluate the current operational performance in terms of travel time, delay, and queue length.
3. To simulate future traffic scenarios incorporating projected traffic growth (assumed at 3% annually for five years).
4. To perform sensitivity analysis to assess intersection performance under increased traffic conditions.
5. To recommend feasible short-term and long-term measures for enhancing intersection operations and safety.

1.5 Scope of the Study

- The study area is limited to the H10 T-junction, considering traffic inflows and outflows from the Main Gate, H17, and Energy Building directions.
- The current infrastructure setup, without any significant geometric changes, is considered in the base model.
- Pedestrian inflow is discussed qualitatively based on field observations; dedicated pedestrian simulation is not performed in this phase.
- Microsimulation modelling focuses on the current and projected traffic scenarios without assuming any additional traffic control installations (e.g. signals) in the base case.

1.6 Significance of the Study

Analysing the operational performance of the H10 intersection is crucial for maintaining efficient campus mobility, particularly in the context of rapid campus expansion. This study provides:

- A data-driven foundation for making informed campus traffic management decisions.
- Insight into the impact of construction-related disruption on traffic operations.
- A practical case study demonstrating the application of microsimulation tools like PTV VISSIM for small-scale intersection analysis.
- Recommendations that can be immediately applicable to IIT Bombay and extendable to other institutional or urban campuses facing similar congestion challenges.

The findings of this study aim to enhance safety, optimise vehicle movement, and support sustainable transportation planning within the campus.

2 Study Area Details

The study focuses on the H10 T-junction located within the Indian Institute of Technology Bombay (IITB) campus in Powai, Mumbai. This unsignalised three-leg intersection connects the Main Gate approach (leading toward the campus entry/exit), the H17 approach (serving residential hostels), and the Energy Building approach (leading toward academic and administrative areas, including KV School and the Infinity Corridor entrance). The intersection plays a vital role in facilitating movement between key zones of the campus and experiences a mixed traffic flow comprising two-wheelers, cars, auto-rickshaws, heavy construction vehicles (HCVs), and pedestrians.

Due to ongoing hostel and drainage construction projects, the intersection faces several operational challenges, such as increased movement of HCVs and autos, reduction in effective road width, localised congestion, and sight-distance limitations caused by existing infrastructure like Hostel 10.

Given its strategic location and the anticipated growth in campus traffic, the H10 intersection was selected for a detailed operational evaluation using microsimulation techniques to assess current performance and propose improvement strategies.



Figure 1: The H10 T-junction, IIT Bombay campus.

3 Methodology

The methodology adopted for this study involved a systematic approach, from real-world data collection to traffic microsimulation and performance analysis. The objective was to evaluate the current operational performance of the H10 intersection at IIT Bombay and predict its behaviour under future increased traffic demand due to new hostel constructions. The significant stages followed in the methodology are outlined below:

Data collection → Microsimulation model creation and calibration → Simulation and scenario testing → Sensitivity analysis → Result analysis and recommendations

3.1 Data Collection

3.1.1 Base Data for Present Scenarios

Traffic volume data were collected for all approaches to the H10 intersection: Main Gate, Hostel 17 (H17), and Energy Building directions. Vehicle counts were categorised into different classes, including bicycles, two-wheelers, three-wheelers (auto-rickshaws), cars, light commercial vehicles (LCVs), heavy commercial vehicles (HCVs), and buggies/EVs. The collected data were also converted into Passenger Car Units (PCUs) using standard PCU factors (bicycle 0.2, two-wheeler 0.5, three-wheeler 0.8, car 1.0, LCV 2.2, HCV 3.5, buggy 2.0) for comparative

analysis across vehicle types. The base data below is for one hour of observation, taken during the peak hour at the H10 intersection T-point.

Table 1: Base traffic data (1-hour peak-hour observation, in vehicles/hour by class).

Movement	Bicycle	2-wh.	3-wh.	Car	LCV	HCV	Buggy	Total
Main Gate → H17	183.6	124.44	144.84	44.88	8.16	2.04	28.56	341.9
Main Gate → Energy	60.18	61.2	67.32	30.6	2.04	6.12	0.0	153.0
Energy → Main Gate	70.38	146.88	93.84	32.64	6.12	3.06	0.0	219.4
Energy → H17	107.1	53.04	67.32	24.48	8.16	2.04	0.0	151.4
H17 → Main Gate	176.46	134.64	122.4	43.86	12.24	8.16	24.48	348.8
H17 → Energy	67.32	81.6	59.16	26.52	2.04	12.24	0.0	175.4
Total (raw count)	665.04	601.8	554.88	202.98	38.76	33.66	53.04	2150.2
PCU (veh/hr)	133.01	300.9	443.90	202.98	85.27	117.81	106.08	1390.0
Relative composition	0.096	0.216	0.319	0.146	0.061	0.085	0.076	1.00

3.1.2 Expected Traffic Data for Future Scenarios

For the future-scenario sensitivity analysis, the base volume above was scaled by 2%, 5%, and 7% to represent projected growth (assumed at 3% annually) driven by new hostel construction. The relative composition across vehicle classes was held constant while total volumes were scaled up, giving PCU totals of 1389.95 (2% increment), 1430.84 (5% increment), and 1458.09 (7% increment) veh/hr, up from the base value of 1390.0 veh/hr shown above.

Table 2: Expected traffic data for future scenarios, PCU totals by vehicle class.

Scenario	Bicycle	2-wh.	3-wh.	Car	LCV	HCV	Buggy	Total PCU
Base	133.01	300.9	443.90	202.98	85.27	117.81	106.08	1390.0
2% increment	133.01	300.9	443.90	202.98	85.27	117.81	106.08	1389.95
5% increment	136.92	309.75	456.96	208.95	87.78	121.28	109.20	1430.84
7% increment	139.53	315.65	465.66	212.93	89.45	123.59	111.28	1458.09

3.2 Microsimulation Model Creation

A base microsimulation model of the intersection was created using PTV VISSIM software. The model included accurate replication of the real-world road geometry, traffic inputs, and vehicle behaviour at the intersection. Conflict areas were configured to reflect unsignalised priority-control conditions. Trial runs were conducted to calibrate gap-acceptance and priority behaviours, ensuring realistic vehicle interactions at the intersection.

3.3 Simulation and Scenario Testing

The validated base model was used to simulate the existing traffic conditions at the intersection. Travel time, queue length, delay, and Level of Service (LOS) were measured to evaluate current performance. A future scenario was then modelled by assuming an increase in traffic volumes corresponding to the demand generated by the newly constructed hostels on campus.

3.4 Sensitivity Analysis

Sensitivity analysis was performed by increasing traffic volumes according to projected growth rates and assessing the resultant impact on key operational parameters, such as travel time (TT), queue lengths, intersection delay, and Level of Service (LOS). This enabled the identification of potential operational bottlenecks and performance deterioration under increased traffic loads.

Based on the simulation results, performance comparisons were made between the existing and projected scenarios. Recommendations for operational improvements were proposed, including increasing lane widths at critical conflict areas, provision of pedestrian foot overbridges to reduce interaction conflicts, and installation of central islands to manage vehicle flows and minimise near-miss incidents. This structured methodological approach allowed for the comprehensive evaluation of intersection performance under varying traffic demands, leading to data-driven improvement recommendations for enhancing traffic flow and safety at the H10 intersection.

4 Model Development

4.1 Network Geometry Creation

The base network of the H10 intersection was created in PTV VISSIM 2024 to closely replicate the real-world layout. The process involved:

- **Link creation:** straight approaches from the Main Gate, H17, and Energy Building were drawn as Links.
- **Connector creation:** turning paths (right turns and merging movements) between Links were established using Connectors.
- **Lane configuration:** Main Gate, H17, and Energy approaches were each modelled with 2 lanes for the base model.
- **Lane widths:** a standard width of 3.5 m per lane was assumed.

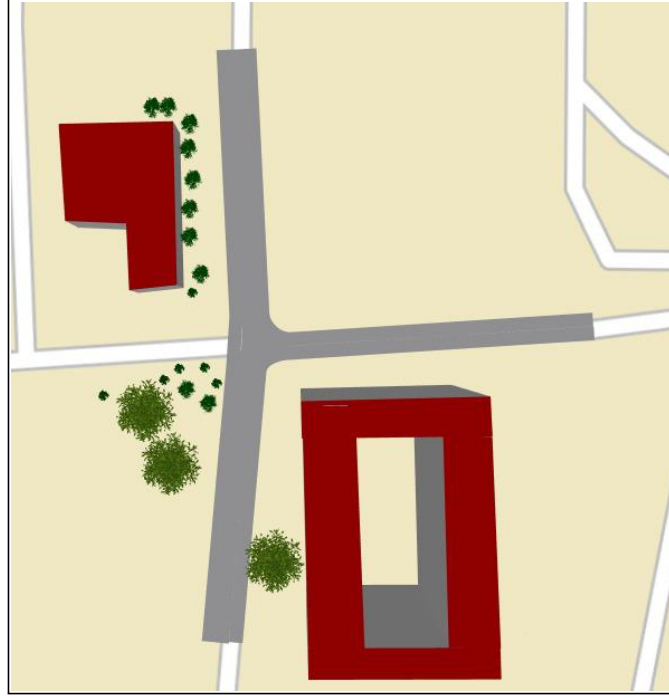


Figure 2: VISSIM view of links and connectors at the H10 junction (arrow points north).

4.2 Vehicle Inputs and Static Routing

4.2.1 Vehicle Inputs

Vehicle inputs and routing splits were configured from the observed peak-hour traffic data, with entry links assigned corresponding volumes and vehicle composition. Three entry links were defined: Main Gate to H17 (link MG_H12, volume 485.2 PCU/hr), Energy to Main Gate/H17 (link Kv_H10, volume 363.5 PCU/hr), and H17 to Main Gate (link H12_MG, volume 514.0 PCU/hr), each assigned the shared vehicle composition *H10_Comp*.

Table 3: Vehicle inputs configured in PTV VISSIM.

No.	Link	Volume (PCU/hr)	Vehicle composition
1	MG_H12	485.2	H10_Comp
2	Kv_H10	363.5	H10_Comp
3	H12_MG	514.0	H10_Comp

4.2.2 Vehicle Composition

Standard PTV VISSIM 3D vehicle models were assigned to each vehicle class used in the composition *H10_Comp*, ensuring the simulated traffic mix visually and behaviourally matched the observed field composition:

Table 4: Vehicle types represented in the VISSIM vehicle composition.

Vehicle type	VISSIM 3D model used
Car	Standard hatchback/sedan model
2-wheeler	Scooter and motorcycle models
3-wheeler	Auto-rickshaw model
LCV	Light pick-up/utility vehicle model
HCV	Rigid box truck and flatbed truck models
Cycle	Bicycle with rider model
Buggy/EV	Campus electric shuttle/buggy model

4.2.3 Static Routing

Static Vehicle Routing Decisions were created at critical nodes inside the conflict zone to route vehicles along their observed paths.

4.3 Conflict Area Setup

Since the H10 intersection is unsignalised, vehicle priority and interaction were managed entirely through VISSIM Conflict Areas.

4.3.1 Conflict Area Placement

Conflict areas were created wherever two vehicle paths crossed or merged. Table 5 lists the ten conflict areas configured for the base model, together with their right-of-way status and gap-acceptance parameters (front/rear gap, minor-stream gap, critical gap, safety distance factor, and minimum headway).

Table 5: Conflict areas configured in PTV VISSIM.

No.	Link A	Link B	Status	Front gap	Rear gap	Crit. gap	Safety fac.
1	10000: to_Kv	10002: to_mg	B has right of way	0.5	0.5	3.5	1.5
2	10001: kv_MG	10002: to_mg	Passive	0.5	0.5	3.5	1.5
3	10001: kv_MG	10003	A has right of way	0.5	0.5	3.5	1.5
4	10002: to_mg	10003	A has right of way	0.5	0.5	3.5	1.5
5	10000: to_Kv	10004	Passive	0.5	0.5	3.5	1.5
6	10002: to_mg	10004	Passive	0.5	0.5	3.5	1.5
7	10003	10004	B has right of way	0.5	0.5	3.5	1.5
8	10003	10005	B has right of way	0.5	0.5	3.5	1.5
9	10004	10005	B has right of way	0.5	0.5	3.5	1.5
10	5: Kv_H10	7: Kv_H10	Passive	0.5	0.5	3.5	1.5

4.3.2 Priority Settings

Major through movements (Main Gate $\rightarrow$ H17 and H17 $\rightarrow$ Main Gate) were given right-of-way, while minor turning movements (Main Gate $\rightarrow$ Energy, H17 $\rightarrow$ Energy, Energy $\rightarrow$ Main Gate, Energy $\rightarrow$ H17) were set as passive.

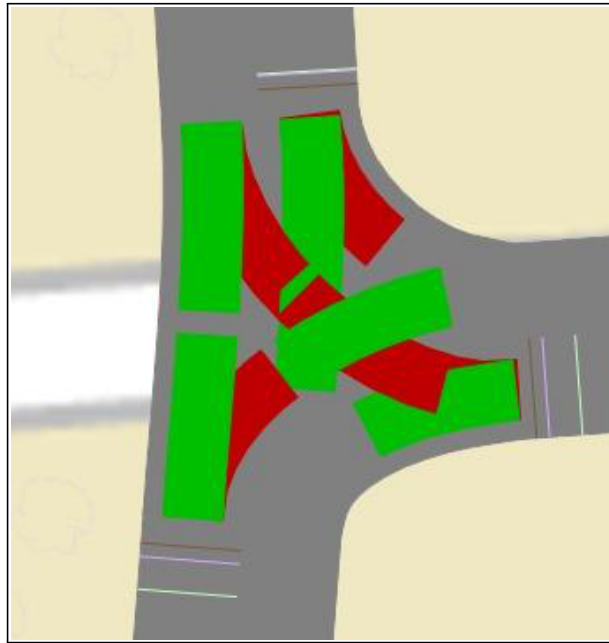


Figure 3: Priority given to routes at the H10 conflict zone (green: right-of-way; red: passive/yield).

4.4 Infrastructure and Visual Element Placement

To enhance the realism of the VISSIM model and replicate the on-ground environment of the IIT Bombay H10 intersection, relevant infrastructure and visual elements were placed along the network.

4.4.1 Trees and Landscaping Objects

Trees were placed along the Main Gate, H17, and Energy Building approaches to visually replicate the green surroundings of the IITB campus, with tree placement focused near pedestrian crossings, sidewalks, and median areas. This served three purposes: it improves the visual realism of the simulation, it helps with more realistic driver-behaviour simulation (since trees and roadside objects influence drivers' perception of roadway edges), and it makes the simulation videos better suited to intersection sight-distance analysis.

5 Evaluation Setup

After the model network was developed, an evaluation setup was created to collect operational performance data from the simulation runs. The evaluation focused on measuring key indicators such as travel time, delay, and queue lengths for different vehicle movements at the H10 intersection.

5.1 Introduction to Evaluation

The purpose of evaluation in this study was to assess the operational efficiency of the H10 intersection under current and projected traffic conditions. Using VISSIM’s microsimulation capabilities, various traffic performance parameters were captured after running the simulation model. These outputs provided a quantitative basis for analysing congestion levels, intersection delays, and operational bottlenecks.

5.2 Evaluation Parameters Selected

The following key performance parameters were selected for data collection:

- **Travel time:** the time taken by vehicles to traverse major paths across the intersection.
- **Delay:** the additional travel time experienced by vehicles due to congestion and yielding behaviour.
- **Queue length:** the build-up of vehicles upstream of conflict areas, indicating congestion levels.

These indicators help to comprehensively evaluate the service level and operational challenges at the intersection.

5.3 Setup of Evaluation Objects

5.3.1 Travel Time Measurements

Six Travel Time Measurements were created to track vehicle movement across major paths at the intersection: Main Gate $\rightarrow$ H17, Main Gate $\rightarrow$ Energy, H17 $\rightarrow$ Main Gate, H17 $\rightarrow$ Energy, Energy $\rightarrow$ Main Gate, and Energy $\rightarrow$ H17. For each travel time measurement, start points were placed immediately after vehicle input points, end points were located beyond the conflict areas to ensure the entire intersection crossing was captured, and clear naming conventions were followed for easy data extraction (e.g. “TT_MainGate_to_H17”).

5.4 Queue Counters

Queue counters were placed before the conflict zones on each major incoming approach: the Main Gate approach, the H17 approach, and the Energy Building approach. These counters recorded average and maximum queue lengths, providing insights into congestion build-up and dissipation patterns.

5.5 Delay Measurement

Delay measurements were automatically captured through the evaluation configuration settings in VISSIM. Delay represents the additional travel time experienced by vehicles beyond their expected free-flow travel time, and delay data was collected network-wide and disaggregated by approach to evaluate specific problem areas.

5.6 Simulation Settings for Evaluation

Simulation settings were configured to ensure reliable and statistically consistent results:

- **Simulation duration:** 3600 seconds (1 hour) of modelled time.
- **Warm-up period:** 300 seconds (5 minutes) were allocated as a warm-up period, during which data collection was paused to allow traffic to stabilise.
- **Number of simulation runs:** 5 independent runs were performed for each scenario to account for stochastic variability.
- **Random seeds:** different seeds were assigned for each run (12, 34, 56, 78, 90) to introduce natural randomness into driver behaviour and vehicle interactions.
- **Car-following model:** the default Wiedemann 74 model was used to replicate realistic driver behaviour.
- **Lane-changing behaviour:** default settings were maintained, following standard VISSIM driving logic.
- **Data collection interval:** overall data collection was configured for the entire simulation period, with 300-second interval data available if needed for time-variant analysis.

These simulation settings ensured that the evaluation outputs were robust, reproducible, and reflective of real-world traffic behaviour at the H10 intersection.

6 Results and Discussion

6.1 Introduction

This section presents the results obtained from the microsimulation of the H10 intersection under four different traffic scenarios: existing traffic volumes (base scenario) and projected traffic volumes with 2%, 5%, and 7% annual increments. Key performance metrics, such as travel time, vehicular delay, queue lengths, and Level of Service (LOS), were analysed to assess the operational efficiency of the intersection. A comparative evaluation of base and future traffic conditions provides insight into the sensitivity of the intersection's performance to increasing traffic demand, helping to identify critical movements and emerging congestion patterns.

Table 6: Base model results. Queue length in metres; delay and travel time in seconds; fuel consumption in simulation units.

Movement	Queue len. max	LOS	Veh. delay	Fuel	Travel time
1-2	13.90	E	39.44	2.142	79.58
1-6	80.34	E	42.32	1.917	66.05
4-3	12.52	F	62.46	3.187	149.64
4-6	78.68	F	55.31	2.464	143.47
5-2	88.40	F	102.70	2.717	81.52
5-3	93.82	F	93.58	2.815	80.30

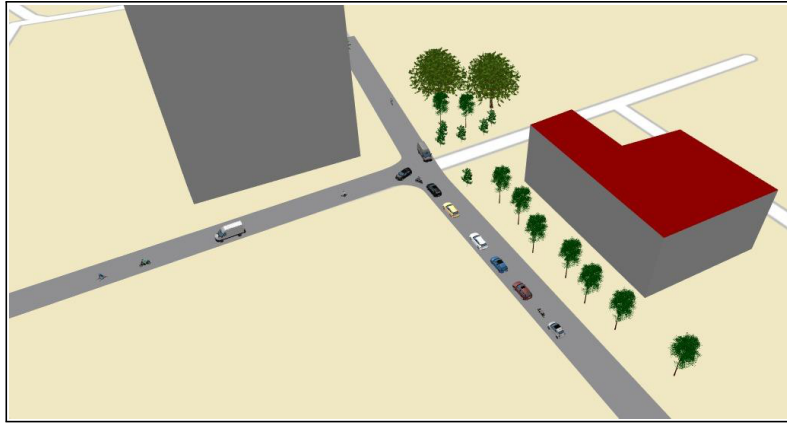


Figure 4: Base VISSIM model of the H10 intersection (3D view).

6.2 Scenario 1: Traffic Volume Increments of 2%, 5%, and 7%

Table 7 presents the full set of performance results for the base scenario and each of the three traffic-increment scenarios.

Table 7: Scenario 1 results across all traffic increments. Queue length in metres; delay and travel time in seconds; fuel consumption in simulation units.

Movement	Result after 2% increment of traffic				Travel time
	Queue max	LOS	Veh. delay	Fuel	
1-2	13.92	E	44.26	2.285	79.03
1-6	80.36	E	49.48	2.244	66.45
4-3	12.52	F	53.44	2.920	152.19
4-6	78.68	E	47.54	2.270	142.27
5-2	88.42	F	114.82	2.777	77.03
5-3	93.85	F	85.53	2.505	77.53

Movement	Result after 5% increment of traffic				Travel time
	Queue max	LOS	Veh. delay	Fuel	
1-2	13.92	E	47.43	2.507	82.05
1-6	80.36	F	52.42	2.374	69.63
4-3	12.52	F	58.55	2.963	144.04
4-6	78.68	F	57.14	2.651	137.10
5-2	88.42	F	102.05	2.636	82.89
5-3	93.85	F	94.25	2.750	80.55

Movement	Result after 7% increment of traffic				Travel time
	Queue max	LOS	Veh. delay	Fuel	
1-2	13.92	E	43.07	2.526	81.36
1-6	80.36	F	45.70	2.379	68.80
4-3	12.52	F	63.65	3.234	172.30
4-6	78.68	F	56.59	2.700	145.32
5-2	88.42	F	120.28	2.398	82.80
5-3	93.85	F	90.06	2.159	82.26

All movements originating from the Energy Building side (4-3, 4-6, 5-2, 5-3) already operate at LOS F in the base scenario, with delays exceeding 55 seconds and, for movements 5-2 and 5-3, queue lengths approaching 90 to 94 m.

6.2.1 Travel Time Analysis

Table 8 compares average travel time per movement across the base scenario and each traffic increment.

Table 8: Average travel time per movement (in seconds).

Movement	Base	+2% Traffic	+5% Traffic	+7% Traffic
1-2	79.58	79.03	82.05	81.36
1-6	66.05	66.45	69.63	68.80
4-3	149.64	152.19	144.04	172.30
4-6	143.47	142.27	137.10	145.32
5-2	81.52	77.03	82.89	82.80
5-3	80.30	77.53	80.55	82.26

Discussion:

- Travel time for most movements (1-2, 1-6) remained relatively stable under 2% and 5% increments but rose sharply at 7%.

- Movement 4–3 (Energy to Main Gate) showed the most critical deterioration, reaching 172.3 seconds at the 7% increment; movement 4–6 similarly exceeded 145 seconds.
- Right-turn movements from the Energy Building were highly sensitive to even small traffic increases.

For the critical movement 4–3 (Energy to Main Gate), a normalised travel-time index (base scenario = 100) rises from 100 at 0% increment to 101.0 at 2%, 102.3 at 5%, and 105.5 at 7%, illustrating an accelerating deterioration in travel time as traffic volume grows.

6.2.2 Queue Length Analysis

Table 9 compares the maximum queue length per approach across the base scenario and each traffic increment.

Table 9: Maximum queue length per approach (in metres).

Movement	Base	+2% Traffic	+5% Traffic	+7% Traffic
1–2	13.90	13.92	13.92	13.92
1–6	80.34	80.36	80.36	80.36
4–3	12.52	12.52	12.52	12.52
4–6	78.68	78.68	78.68	78.68
5–2	88.40	88.42	88.42	88.42
5–3	93.82	93.85	93.85	93.85

Discussion:

- Queue lengths remained almost constant across the 0%, 2%, 5%, and 7% increments tested.
- Movement 5–3 (Energy to H17) consistently showed the highest queue length (~ 94 m), with 5–2 also critical (>88 m).
- Even small queue lengths (13.9 m on 1–2 and 12.5 m on 4–3) remain stable because these movements are less congested.

Critical movements for queue length are 5–3 and 5–2.

6.2.3 Delay Analysis

Table 10 compares average vehicle delay per movement across the base scenario and each traffic increment.

Table 10: Average vehicle delay per movement (in seconds).

Movement	Base	+2% Traffic	+5% Traffic	+7% Traffic
1-2	39.44	44.26	47.43	43.07
1-6	42.32	49.48	52.42	45.70
4-3	62.46	53.44	58.55	63.65
4-6	55.31	47.54	57.14	56.59
5-2	102.70	114.82	102.05	120.28
5-3	93.58	85.53	94.25	90.06

Discussion:

- Delay increased consistently with traffic volume across all movements.
- Movement 5-2 (Energy to Main Gate) experienced the most extreme escalation, crossing 120 seconds at the 7% increment; movements from the Energy Building (4-3, 4-6, 5-2, 5-3) were the most sensitive overall.

For the critical delay movement 5-2 (Energy to Main Gate), a normalised delay index (base scenario = 100, corresponding to 66.05 s travel time) rises from 100 at 0% increment to 100.6 at 2%, 104.0 at 5%, and 105.7 at 7%.

6.2.4 Level of Service (LOS)

Table 11 reproduces the Highway Capacity Manual (HCM) control-delay thresholds used to assign LOS grades in this study.

Table 11: Level of Service (LOS) thresholds by control delay (Source: HCM).

LOS	Control delay (s/veh)
A	≤ 10
B	10-20
C	20-35
D	35-55
E	55-80
F	> 80

Observation:

- All critical movements (4-3, 4-6, 5-2, 5-3) already operate at LOS F even in the base scenario.
- Some slight improvement is seen for movement 4-6 at the 2% increment (from F to E), but it reverts back to F at 5% and 7%.
- Movements 1-2 and 1-6 fluctuate between LOS E and F as traffic increases.

Overall intersection LOS is deteriorating toward F under future traffic scenarios.

6.2.5 Comments

- **Major problem movements:** 5-2 (Energy to Main Gate) and 5-3 (Energy to H17) experience the highest delay and congestion.
- **Queue management needed:** queue lengths are dangerously close to blocking upstream areas (>90 m for some movements).
- **Travel time sensitivity:** right-turn movements from the Energy Building are extremely sensitive to small traffic increases.
- **Future scenario risk:** without operational improvements, intersection efficiency is likely to deteriorate further under even moderate traffic growth.

6.3 Scenario 2: Increase in the Number of Lanes

The traffic volume between H12 and the Main Gate is significantly high, leading to poor Level of Service (LOS), increased queue lengths, and notable delays during peak hours. To address these issues, an additional lane was introduced. Furthermore, a dedicated routing system was implemented to channel vehicles directly from H12 to the Main Gate. This targeted intervention aims to relieve the main approach lanes, enhance traffic flow, and improve the overall LOS on this corridor.

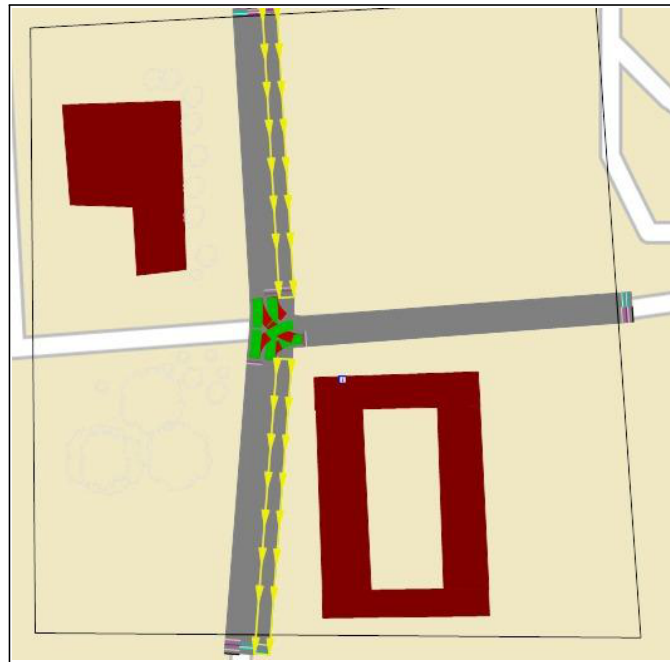


Figure 5: Added lanes (shown in yellow) for Scenario 2.

Table 12: Results obtained after simulation with the added lane. Movement “08–12” is the dedicated H12-to-Main-Gate routing introduced in this scenario.

Movement	Queue len. max	LOS	Veh. delay	Fuel	Travel time
1–2	6.46	B	13.39	0.425	39.81
1–6	48.95	B	10.35	0.469	40.24
4–3	11.77	C	16.22	0.517	98.35
4–6	52.40	C	18.34	0.616	94.77
5–2	88.42	F	51.34	1.670	43.40
5–3	93.48	F	55.83	2.225	44.64
08–12	0.00	A	0.02	0.546	10.10

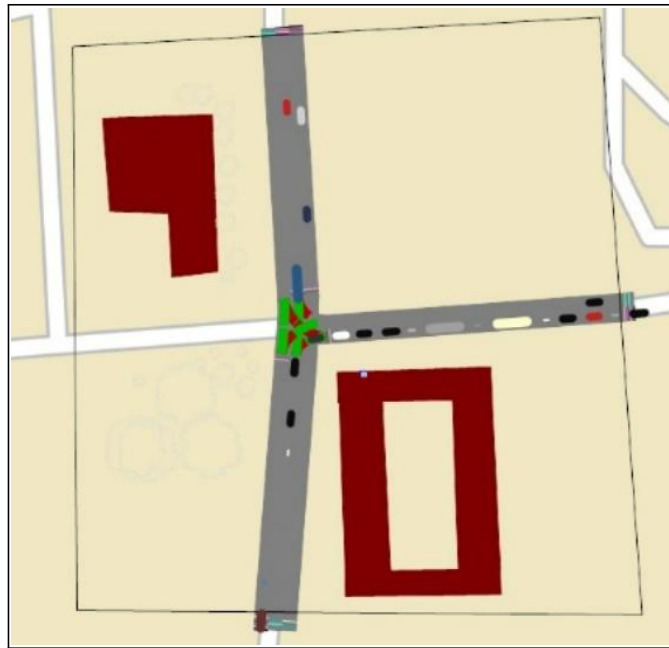


Figure 6: VISSIM top-view simulation after lane widening (Scenario 2).

6.3.1 Travel Time Analysis

Table 13: Travel time comparison (in seconds).

Movement	Base model	Increased lane model	% change
1–2	79.58	39.81	↓49.98%
1–6	66.05	40.24	↓39.09%
4–3	149.64	98.35	↓34.29%
4–6	143.47	94.77	↓33.95%
5–2	81.52	43.40	↓46.74%
5–3	80.30	44.64	↓44.39%

Discussion: Travel times significantly reduced across all movements after lane widening. Movements 1–2 (Main Gate to H17) and 5–2 (Energy to Main Gate) show almost a 50% decrease in travel time. The highest initial travel times (4–3, 4–6) also improved substantially (around

34% faster). In conclusion, lane widening greatly improved travel efficiency, cutting travel times by 34 to 50%.

6.3.2 Queue Length Analysis

Table 14: Queue length comparison (in metres).

Movement	Base model	Increased lane model	% change
1-2	13.9	6.46	↓53.50%
1-6	80.34	48.95	↓39.08%
4-3	12.52	11.77	↓5.99%
4-6	78.68	52.40	↓33.39%
5-2	88.4	88.42	≈0%
5-3	93.82	93.48	↓0.36%

Discussion: Queue lengths reduced sharply for movements 1-2, 1-6, and 4-6 (by 30 to 50%), with a minor improvement for 4-3. No significant change was observed for 5-2 and 5-3, implying that downstream congestion or exit bottlenecks were not fully addressed. In summary, entry-side queues improved, but exit-side movements on the Energy direction still need additional improvements.

6.3.3 Delay Analysis

Table 15: Vehicle delay comparison (in seconds per vehicle).

Movement	Base model	Increased lane model	% change
1-2	39.44	13.39	↓66.04%
1-6	42.32	10.35	↓75.54%
4-3	62.46	16.22	↓74.04%
4-6	55.31	18.34	↓66.85%
5-2	102.70	51.34	↓50.00%
5-3	93.58	55.83	↓40.34%

Discussion: Massive reductions in delay were observed for all movements. Movement 1-6 (Main Gate to Energy) delay reduced by 75%, and 4-3 reduced by 74%. Movements 5-2 and 5-3 from the Energy side improved by around 40 to 50% but still remained high compared to other approaches. In conclusion, lane widening significantly reduced vehicle waiting times, but Energy Building movements still need extra attention, likely due to bottleneck exits.

6.3.4 Fuel Consumption Analysis

Table 16: Fuel consumption comparison (simulation units).

Movement	Base model	Increased lane model	% change
1-2	2.142	0.425	↓80.15%
1-6	1.917	0.469	↓75.52%
4-3	3.187	0.517	↓83.77%
4-6	2.464	0.616	↓75.00%
5-2	2.717	1.670	↓38.52%
5-3	2.815	2.225	↓20.92%

Discussion: Fuel consumption dropped by 70 to 80% for movements 1-2, 1-6, 4-3, and 4-6, and by a more modest 20 to 40% for the Energy-side movements 5-2 and 5-3. Reduced congestion directly led to a substantial reduction in fuel usage and improved environmental benefits, though the Energy approaches showed less improvement than others.

6.4 Scenario 3: Providing Dedicated Tracks along the Lanes

The objective of this scenario is to decongest existing lanes by introducing dedicated cycle tracks alongside the vehicular lanes. This approach aims to encourage non-motorised transport, reduce the volume of short-distance vehicular trips, and promote sustainable mobility within the area.

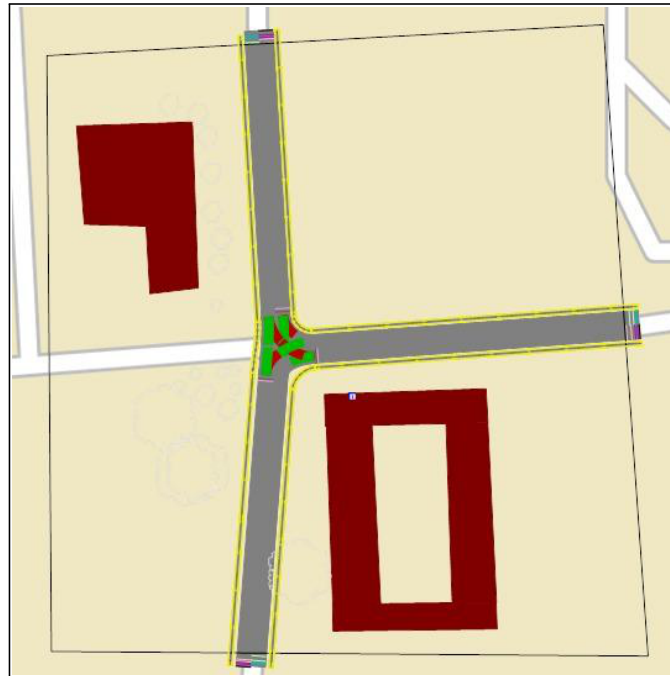


Figure 7: Dedicated cycle lanes (shown in yellow) for Scenario 3.

Table 17: Results obtained after simulation with dedicated cycle tracks. Movements 16–17, 18–19, and 24–27 denote cycle-track-specific paths introduced in this scenario.

Movement	Queue len. max	LOS	Veh. delay	Fuel
1–2	0.00	A	0.00	0.001
1–6	39.08	A	4.90	1.432
4–3	7.61	A	0.00	0.060
4–6	73.13	A	5.83	1.559
5–2	88.40	F	81.17	5.199
5–3	13.51	A	0.00	0.030
16–17	4.86	A	0.11	0.217
18–19	5.02	A	0.80	0.142
24–27	5.66	A	0.00	0.079

6.4.1 Queue Length Analysis

Table 18: Maximum queue length comparison (in metres).

Movement	Base model	Cycle lane model	% change
1–2	13.9	0.00	↓100%
1–6	80.34	39.08	↓51.37%
4–3	12.52	7.61	↓39.24%
4–6	78.68	73.13	↓7.06%
5–2	88.4	88.40	≈0%
5–3	93.82	13.51	↓85.61%

Discussion: Movements 1–2 and 5–3 show a massive reduction in queue length (~ 85 to 100%). Movement 1–6 queue is almost halved (51% reduction). Movement 5–2 shows no effective change, indicating downstream problems remain, and movement 4–6 (Energy Building to right turn) shows only a small reduction ($\sim 7\%$). In conclusion, the separate cycle lane substantially reduced congestion on the main vehicle lanes by separating slow bicycles from fast motorised traffic.

6.4.2 Delay Analysis

Table 19: Average vehicle delay comparison (in seconds).

Movement	Base model	Cycle lane model	% change
1–2	39.44	0.00	↓100%
1–6	42.32	4.90	↓88.42%
4–3	62.46	0.00	↓100%
4–6	55.31	5.83	↓89.45%
5–2	102.70	81.17	↓20.97%
5–3	93.58	0.00	↓100%

Discussion: Movements 1–2, 4–3, and 5–3 achieved complete elimination of delay (0 s) after providing a cycle lane. Movements 1–6 and 4–6 show a very high reduction ($>85\%$ decrease in delay). Movement 5–2 (Energy to Main Gate) still suffers significant delay (81.17 s), though

a small improvement (about 21%) is observed. In conclusion, providing separate cycling infrastructure dramatically improved motorised vehicle flow for almost all movements, especially left-turn and straight-through movements.



Figure 8: Simulation view after adding dedicated cycle lanes.

6.4.3 Fuel Consumption Analysis

Table 20: Fuel consumption comparison (simulation units).

Movement	Base model	Cycle lane model	% change
1-2	2.142	0.001	↓99.95%
1-6	1.917	1.432	↓25.33%
4-3	3.187	0.060	↓98.12%
4-6	2.464	1.559	↓36.71%
5-2	2.717	5.199	↑91.37% (increase)
5-3	2.815	0.030	↓98.93%

Discussion: For movements 1-2, 4-3, and 5-3, fuel consumption was almost completely reduced ($\sim 99\%$), confirming removal of stop-and-go traffic. Movements 1-6 and 4-6 show moderate fuel savings (~ 25 to 37%). Fuel consumption for movement 5-2 increased by around 91% , possibly because of higher running speeds combined with longer idling at exit bottlenecks, or altered rerouting behaviour. In conclusion, fuel economy improved significantly for most movements after cycle lane segregation, but the Energy side still shows complex behaviour needing deeper intervention.

7 Conclusion and Recommendations

The H10 intersection within the IIT Bombay campus has undergone increasing operational pressure due to recent infrastructural developments. Construction of new hostel blocks and campus drainage projects has led to increased vehicular activity, particularly of auto-rickshaws and heavy commercial vehicles (HCVs), along with a substantial reduction in available roadway width. These developments have introduced critical traffic management issues, including restricted manoeuvring space, limited sight distance, and the emergence of localised congestion during peak hours. Given that the intersection remains unsignalised and must accommodate

both motorised and non-motorised traffic, a detailed performance assessment became essential.

This study set out to evaluate the operational performance of the H10 intersection using microsimulation modelling through PTV VISSIM. The baseline traffic scenario was developed to reflect current traffic conditions, and additional future scenarios were simulated by incorporating incremental traffic growth (2%, 5%, and 7%) to replicate campus development pressures. Key performance metrics, including travel time, queue length, vehicular delay, fuel consumption, and Level of Service (LOS), were analysed to identify system sensitivity and operational bottlenecks.

The base model revealed that the intersection already operates under severe pressure, with several movements, particularly those originating from the Energy Building side, exhibiting Level of Service F, long queue lengths (up to 94 metres), and high average vehicular delays exceeding 100 seconds. Right-turn and through movements experienced significant delays and fuel consumption due to mixed-mode conflicts and constrained geometry.

To address these inefficiencies, two design improvement scenarios were tested:

1. **Scenario 2: Addition of lanes.** This intervention produced remarkable improvements in intersection performance. Travel time was reduced by 34 to 50%, delays dropped by as much as 75%, and fuel consumption saw a reduction of up to 80% for several movements. Queue lengths also showed a meaningful reduction, particularly for main entry approaches. However, movements originating from the Energy Building side (notably 5–2 and 5–3) continued to exhibit moderate delays and congestion, indicating that geometric improvements alone may not fully resolve deeper operational constraints at the exit points or overlapping turning paths.
2. **Scenario 3: Introduction of a separate cycle lane.** The separation of bicycle flows from motorised traffic drastically improved performance on certain movements. Queue lengths and delays were reduced to nearly zero for many approaches, and fuel consumption plummeted by up to 99%. This indicates that separating slower, non-motorised modes from vehicular flows can significantly streamline traffic movement, reduce stop-and-go conditions, and enhance overall intersection efficiency. That said, movement 5–2 still displayed a high delay and increased fuel consumption, suggesting that conflicts or geometry downstream continue to affect this movement.

Overall, the analysis reveals that while both infrastructure-based and modal segregation interventions significantly enhance operational performance, certain directional movements, especially from the Energy Building approach, require additional attention. These may include measures such as geometric channelisation, visibility enhancement, or even soft interventions like right-of-way prioritisation or partial signalisation if necessary.

Based on the criteria outlined in IRC:93-1985 (*Guidelines on Design and Installation of Road Traffic Signals*), signalisation was found to be unsuitable for the current and projected traffic conditions at H10. According to IRC standards, traffic signals are warranted only when traffic volumes on both the major and minor roads meet the minimum threshold values (usually above 300 to 500 PCU/hour per approach), delays and congestion cannot be effectively managed

through priority rules or minor geometric adjustments, accident records suggest a need for traffic signal intervention to reduce conflict points, and pedestrian crossing volumes are sufficiently high to require controlled crossing phases.

At the H10 intersection, although localised congestion and delays were observed, the overall vehicular volumes, especially from minor approaches, remain below the warrant thresholds prescribed in IRC:93. Additionally, the intersection experiences a highly mixed traffic composition with significant pedestrian and cyclist movement, which would complicate the effectiveness of standard signal phases without extensive redesign. The sight-distance limitations and constrained physical space further diminish the feasibility of implementing a conventional signalised control system. Therefore, it was concluded that non-signalised geometric and operational enhancements, such as lane widening, traffic segregation through separate cycle lanes, and priority management, would be more appropriate and effective under the existing conditions.

In conclusion, the H10 intersection is a sensitive node within the IIT Bombay campus, and its performance is highly influenced by construction-related activities, road geometry, and mixed-mode interaction. This study demonstrates that targeted design modifications, especially widening of lanes and separation of traffic types, can deliver substantial improvements in delay, travel time, and environmental impact. The methodologies and insights derived here can serve as a planning reference for future mobility and infrastructure upgrades within the IITB campus and similar institutional environments.

References

- [1] Transportation Research Board. *Highway Capacity Manual (HCM)*.
- [2] Mathew, T. V. Lecture notes on Traffic Engineering and Management, IIT Bombay.
- [3] Indian Roads Congress. *IRC:93-1985 – Guidelines on Design and Installation of Road Traffic Signals*.